TEXORA · LECTURE SHEET

Market Research and Product Development

Finding out what buyers actually want before committing fabric and capacity: the research process, primary and secondary data, sampling and questionnaires, SWOT and PESTLE analysis, and the stage-by-stage development of a textile or apparel product from idea to bulk production.

TE-3109 · SUPPLY CHAIN MANAGEMENT IN TEXTILES · TOPIC 10 · LEVEL 3 TERM 1

1. Why research comes before production

Textile production commits money early and recovers it late: fabric is bought, dyed and cut weeks before a garment is sold. A wrong decision — the wrong colour, fibre, price point or market — is therefore discovered only after most of its cost is sunk. **Market research** is the systematic collection and analysis of information about buyers, consumers, competitors and the market environment, undertaken to reduce that risk. **Product development** is its partner process: converting what research learns into a physical product that can be costed, sampled, approved and produced in bulk. In export apparel, the "customer" is two-layered — the **buyer** (retailer or brand placing the order) and the **end consumer** — and research must understand both: the buyer transmits, filters and sometimes distorts the consumer's signal.

2. The market research process

STEP	WHAT HAPPENS	TYPICAL PITFALL
1. Define the problem	State the decision and the information needed ("Will buyers pay Tk 100 more for organic cotton in this range?")	Researching a vague question and getting unusable answers
2. Design the research	Choose exploratory, descriptive or causal design; decide data sources, methods and sample	Copying a previous study that answered a different question
3. Collect the data	Fieldwork: surveys, interviews, observation, experiments; desk work for secondary data	Leading questions, careless recording, unrepresentative respondents

4. Analyse	Clean, tabulate and interpret; estimate market size; test relationships	Reporting averages that hide the segment that matters
5. Report and decide	Present findings tied to the original decision, with costs and risks	A long report with no recommendation
6. Follow up	Compare outcomes with forecasts; feed learning into the next study	Filing the report and repeating the same errors

Exploratory research clarifies an ill-defined problem (interviews with buyers about why orders shifted). **Descriptive** research measures ("what share of buyers source basic tees from Bangladesh?"). **Causal** research tests cause and effect ("did the price cut cause the sales rise, or the season?"), normally needing controlled comparison. Most factory-level studies are exploratory then descriptive; confusing the two — treating a few interviews as measurement — is a classic error.

3. Primary and secondary data

Secondary data already exist: trade statistics (for Bangladesh, Export Promotion Bureau figures and BGMEA/BKMEA publications), customs and import data of destination markets, industry journals, competitors' catalogues and price lists, previous seasons' internal sales and rejection records, and published consumer research. Secondary data are cheap and fast, so research always starts there — but they were collected for another purpose, may be out of date, and their definitions (what counts as "knitwear exports") may not match the study's.

Primary data are collected for the study itself: **surveys** (questionnaires to buyers or consumers), **interviews** (in-depth, with merchandisers or retail buyers), **focus groups** (guided discussion, useful for reactions to new fabrics or styles), **observation** (watching what shoppers actually pick up, which often contradicts what they say), and **experiments** (test displays or price trials in a few stores). Primary data fit the question exactly but cost time and money and can be ruined by poor sampling or poor questions — the subjects of the next two sections.

4. Sampling and market-size estimation

Studying every buyer or consumer is usually impossible, so researchers study a **sample** and generalise to the **population**. In **probability sampling** every member has a known chance of selection — simple random, stratified (sample drawn proportionally from each stratum, such as buyer size bands) or cluster sampling — and results can be projected with measurable error. In **non-probability sampling** — convenience or judgement sampling — selection is by availability or expertise; it suits exploratory work but cannot support precise projection. Sample size trades cost against precision: larger samples

reduce sampling error, with diminishing returns, and a biased small sample is worse than useless because it looks precise while being wrong.

WORKED EXAMPLE — SAMPLING FRACTION AND MARKET SIZE

A merchandising team lists 8,000 retail outlets in a target city (the population) and surveys 400 of them. Sampling fraction = 400 ÷ 8,000 = **5 %**, i.e. 1 outlet in 20.

Market size from survey and census figures: city population 20,00,000; the target age-and-income segment is 25 % of it, giving 20,00,000 × 0.25 = **5,00,000** potential consumers. Survey evidence suggests 40 % of them buy the product type: 5,00,000 × 0.40 = **2,00,000** buyers. Each buys on average 2 shirts a year at an average price of Tk 600: annual market = 2,00,000 × 2 × 600 = **Tk 24,00,00,000 (Tk 24 crore)**, or 4,00,000 shirts a year. Every figure in the chain can now be challenged and refined separately — that transparency is the point of showing the working.

Market-size estimates built this way — population × segment share × penetration × frequency × price — are called **chain-ratio estimates**. Their honesty lies in exposing each assumption: if the 40 % buying rate is really 30 %, the market is Tk 18 crore, not 24, and capacity should be planned against the range, not the single number.

5. The questionnaire

A questionnaire converts research questions into answerable items. Good practice: ask one thing per question; use the respondent's vocabulary (a consumer does not think in GSM); put simple, factual questions first and sensitive ones (price paid, income) late; offer balanced response scales; avoid leading wording ("Don't you agree this fabric feels premium?") and double questions ("Is the fabric soft and durable?"); include screening questions so only the target group answers; and **pilot-test** on a small group before fieldwork, because a question misunderstood by five pilot respondents will be misunderstood by five thousand respondents. Closed questions (fixed choices) analyse easily; a few open questions ("What one change would make you buy more?") catch what the designer did not think to ask.

6. Analysing the environment: SWOT and PESTLE

TOOL	LOOKS AT	ELEMENTS	APPAREL EXAMPLE
SWOT	Internal strengths and weaknesses; external opportunities and threats	S, W (inside the firm) · O, T (outside)	S: strong compliance record. W: no design studio. O: buyers diversifying sourcing. T: rising minimum wages and rival-country duty advantages

PESTLE	The macro-environment	Political, Economic, Social, Technological, Legal, Environmental	Economic: exchange-rate movement changes export competitiveness. Environmental/Legal: wastewater rules raise dyeing costs. Technological: digital printing shortens sampling time

Used together, PESTLE scans the outside world systematically and SWOT organises what it means for this firm. The discipline is in the matching: a strength only matters against an opportunity it can capture (a compliance record matters because buyers are re-sourcing), and a threat only bites through a weakness it can exploit. A SWOT list with no matching and no actions is decoration, not analysis.

7. Product development for textile and apparel

Product development converts an idea into a bulk-producible, costed, buyer-approved product. For an export garment the stages run as follows, with a **gate** after each stage at which the product can be revised or stopped — stopping early is cheap; stopping after bulk fabric is bought is not.

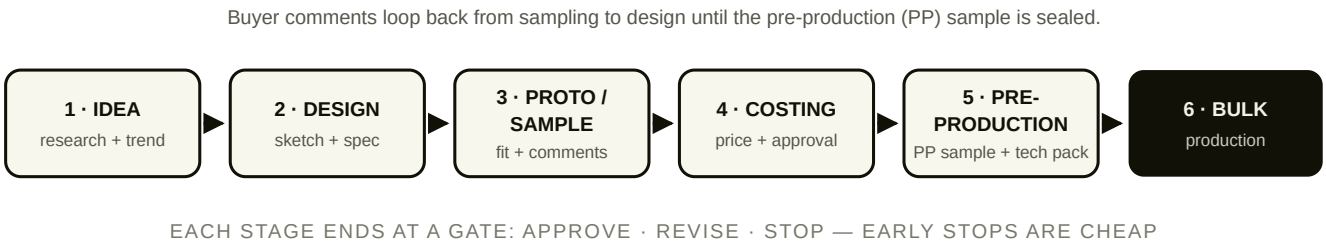


Figure 1 — Product development stages for an apparel product, from researched idea to bulk production. The sampling loop (comments, revised samples) repeats inside stages 2–5 until the buyer seals the pre-production sample.

Idea and design. Ideas come from research, trend forecasting, buyer briefs and fabric development. Design translates the idea into sketches, fabric and trim selection, colourways and a specification. **Prototype and sampling.** The pattern is made and the first proto sample sewn; buyer comments on fit, fabric hand, colour and workmanship drive revised samples — fit samples, size-set samples and salesman samples each serve a distinct approval purpose. **Costing.** The product is costed in full — fabric consumption with wastage, trims, cut-and-make, washing or finishing, testing, packing, freight basis and margin — and the price negotiated against the buyer's target. A style that cannot meet its target price returns to design for engineering (cheaper construction, revised fabric) or is dropped at the gate. **Pre-production.** The approved style is frozen in the **tech pack** and a pre-production (PP) sample is made exactly as bulk will be made, on bulk fabric, and sealed as the production standard.

The **tech pack** is the product's complete instruction set: technical drawings, bill of materials (fabric, trims, labels), measurement chart with tolerances, construction and stitch details, colour and print

specifications, packing and labelling requirements. Every later dispute — a shade band, a measurement failure — is judged against the tech pack, which is why development and quality teams treat its accuracy as sacred. **Buyer requirements** overlay the whole process: restricted-substance and testing standards, labelling and fibre-content rules of the destination market, packaging and carton rules, delivery windows and the buyer's sampling calendar, which is unforgiving — a sample arriving after the buyer's range-selection meeting is a product developed for no one.

8. Linking research, development and the supply chain

The three processes of this topic close the loop of the whole course. Research estimates demand and price sensitivity (Topic 7's elasticity at work); production economics (Topic 6) tests whether the volumes research predicts clear the break-even point; development converts the chosen product into a costed, specified, approved item; project management (Topic 8) plans special undertakings such as a new range launch; law (Topic 9) bounds every claim, label and employment decision; and the supply chain then executes — sourcing, production and delivery against the tech pack and the delivery date. Factories that research first and develop in stages waste less fabric, quote prices they can honour, and present buyers with products the buyers' own customers have, in effect, already chosen.

Glossary

Market research — systematic collection and analysis of market information for a decision. **Primary data** — data collected for the study itself. **Secondary data** — existing data collected for another purpose. **Population** — the whole group to be generalised about. **Sample** — the studied subset of the population. **Stratified sampling** — sampling proportionally from defined subgroups. **Chain-ratio estimate** — market size built by multiplying a chain of stated proportions. **SWOT** — strengths, weaknesses, opportunities, threats analysis. **PESTLE** — political, economic, social, technological, legal and environmental scan. **Tech pack** — the complete technical specification document for a product. **PP sample** — pre-production sample made as bulk will be made and sealed as the standard. **Stage gate** — a decision point where a developing product is approved, revised or stopped.

Key points

• Research exists to reduce the cost of being wrong: define the problem first, then design, collect, analyse, report and follow up.

• Start with cheap secondary data; collect primary data by survey, interview, focus group, observation or experiment when the question demands it.

• Probability sampling supports projection; biased samples mislead. Worked example: 400 of 8,000 outlets is a 5 % sampling fraction, and the chain-ratio market estimate is Tk 24 crore (4,00,000 shirts)

on its stated assumptions.

• Questionnaires ask one clear thing at a time, avoid leading wording, and are pilot-tested.

• PESTLE scans the macro-environment; SWOT matches internal strengths and weaknesses to external opportunities and threats — value lies in the matching.

• Apparel development runs idea → design → proto/sample → costing → pre-production → bulk, gated at each stage, frozen in the tech pack and sealed with the PP sample.

Practice

1. A survey covers 300 factories from a listed population of 6,000. What is the sampling fraction, expressed as a percentage and as "1 in N"?
2. Estimate a market: population 10,00,000; target segment 30 %; buyers among them 50 %; purchases 3 items a year at Tk 450 each. Show the full chain-ratio working and the market value.
3. Rewrite this faulty questionnaire item into two good questions: "Do you agree that our soft and durable fabric is good value?" State what was wrong with the original.
4. Classify each source as primary or secondary: (a) EPB export statistics; (b) your own interviews with five buyers; (c) last season's factory rejection records; (d) a focus group reacting to a new fabric. Justify (c) carefully.
5. Build a mini-SWOT for a small woven factory with excellent quality, no washing plant, a new buyer inquiry for washed garments, and a rival opening a modern wash plant nearby. Match one strength or weakness to the opportunity or threat.
6. A buyer's target price is below your costed price for a new style at the costing gate. List the stage-gate options and the supply chain consequence of choosing the wrong one.

Answer key

1. 300 ÷ 6,000 = 0.05 = 5 %, i.e. 1 in 20 factories.
2. Target segment = 10,00,000 × 0.30 = 3,00,000 people. Buyers = 3,00,000 × 0.50 = 1,50,000. Units = 1,50,000 × 3 = 4,50,000 items a year. Market value = 4,50,000 × 450 = Tk 20,25,00,000 (Tk 20.25 crore) a year.
3. Example rewrite: (i) "How would you rate the softness of this fabric?" (balanced scale: very soft … very harsh). (ii) "How would you rate its durability?" plus a separate value question, "At Tk X, how likely are you to buy?" Faults in the original: it is leading ("Do you agree"), double-barrelled (soft *and* durable in one item), and mixes quality with value for money, so no answer can be interpreted.
4. (a) Secondary — published for general use. (b) Primary — collected for this study. (c) Secondary — the records exist for operational purposes and are being re-used for research, even though they are

internal; classification depends on purpose of original collection, not location. (d) Primary — generated by the study.

5. Strength: excellent quality record. Weakness: no washing plant. Opportunity: buyer inquiry for washed garments. Threat: rival's new wash plant capturing washed-garment orders. Matching: the weakness blocks the opportunity directly — options are subcontract washing under strict quality control (using the strength to manage it), invest via a project plan, or decline the inquiry before the rival takes the buyer relationship entirely; the threat makes delay costly.
6. Options at the gate: re-engineer the product (fabric, construction, trims) to meet the target without breaking the specification; negotiate (price, quantity, delivery or specification changes); or stop the style before bulk commitments. The wrong choice — accepting below cost to "keep the line busy" without a contribution case, or quietly cheapening materials — converts a development-stage problem into bulk-stage losses, claims and audit risk across the whole supply chain.
